

SE LAB 3 : Architecture Selection and Justification

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Architecture Selection

We chose Microservices Architecture for the Multi-Vendor Artisan E-Commerce Marketplace.

Reason 1 – Independent Services

The marketplace contains different functions such as vendor management, product catalogue management, order processing and payment processing. Microservices allows these functions to be separated into independent services, making the system modular and easier to maintain.

Reason 2 – Scalability

A multi-vendor marketplace can have many vendors and customers accessing the system simultaneously. Individual services such as the Product Catalogue or Order Management Service can be scaled independently according to demand.

Security Advantage

Separating payment processing from other marketplace services helps isolate sensitive payment operations. Access to payment functionality can be restricted to authorized service requests instead of allowing every component to directly access payment information.

Performance Benefit

Frequently used services such as the Product Catalogue can be scaled independently when the number of product searches increases. This allows the system to handle high workloads without requiring every service to scale at the same time.